

Data Quality Report

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Executive Summary

Metric	Value
Total Datasets	4
Datasets Passed Validation	1
Datasets with Issues	3
Total Validation Issues	4
Average Quality Score	100.00%

Data Validation Results

movies - Status: **FAIL**

Total Rows	9742
Total Columns	3
Validation Checks Passed	3/4

Check	Status	Details
Schema	PASS	All required columns present
Missing Values	PASS	No missing values detected
Duplicates	PASS	No duplicate rows detected
Data Types	FAIL	Data type mismatches found in 2 columns

ratings - Status: **PASS**

Total Rows	100836
Total Columns	4
Validation Checks Passed	5/5

Check	Status	Details
Schema	PASS	All required columns present
Missing Values	PASS	No missing values detected
Duplicates	PASS	No duplicate rows detected
Data Types	PASS	All data types match schema
Constraints	PASS	All constraints satisfied

links - Status: **FAIL**

Total Rows	9742
Total Columns	3
Validation Checks Passed	2/4

Check	Status	Details
Schema	PASS	All required columns present
Missing Values	FAIL	Found 8 null values across columns
Duplicates	PASS	No duplicate rows detected
Data Types	FAIL	Data type mismatches found in 1 columns

tags - Status: **FAIL**

Total Rows	3683
Total Columns	4
Validation Checks Passed	3/4

Check	Status	Details
Schema	PASS	All required columns present
Missing Values	PASS	No missing values detected
Duplicates	PASS	No duplicate rows detected
Data Types	FAIL	Data type mismatches found in 1 columns

Data Profiling & Statistics

movies

Column	movieId
Data Type	int64
Non-Null Count	9742
Null %	0.00%
Unique Values	9742
Min	1.00
Max	193609.00
Mean	42200.35
Std Dev	52160.49

Column	title
Data Type	str
Non-Null Count	9742
Null %	0.00%
Unique Values	9737

Column	genres
Data Type	str
Non-Null Count	9742
Null %	0.00%
Unique Values	951

ratings

Column	userId
Data Type	int64
Non-Null Count	100836
Null %	0.00%
Unique Values	610
Min	1.00
Max	610.00
Mean	326.13
Std Dev	182.62

Column	movieId
Data Type	int64

Non-Null Count	100836
Null %	0.00%
Unique Values	9724
Min	1.00
Max	193609.00
Mean	19435.30
Std Dev	35530.99

Column	rating
Data Type	float64
Non-Null Count	100836
Null %	0.00%
Unique Values	10
Min	0.50
Max	5.00
Mean	3.50
Std Dev	1.04

Column	timestamp
Data Type	int64
Non-Null Count	100836
Null %	0.00%
Unique Values	85043
Min	828124615.00
Max	1537799250.00
Mean	1205946087.37
Std Dev	216261036.00

links

Column	movieId
Data Type	int64
Non-Null Count	9742
Null %	0.00%
Unique Values	9742
Min	1.00
Max	193609.00
Mean	42200.35

Std Dev	52160.49
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Column	imdbId
Data Type	int64
Non-Null Count	9742
Null %	0.00%
Unique Values	9742
Min	417.00
Max	8391976.00
Mean	677183.90
Std Dev	1107227.58

Column	tmdbId
Data Type	float64
Non-Null Count	9734
Null %	0.08%
Unique Values	9733
Min	2.00
Max	525662.00
Mean	55162.12
Std Dev	93653.48

tags

Column	userId
Data Type	int64
Non-Null Count	3683
Null %	0.00%
Unique Values	58
Min	2.00
Max	610.00
Mean	431.15
Std Dev	158.47

Column	movieId
Data Type	int64
Non-Null Count	3683
Null %	0.00%
Unique Values	1572

Min	1.00
Max	193565.00
Mean	27252.01
Std Dev	43490.56

Column	tag
Data Type	str
Non-Null Count	3683
Null %	0.00%
Unique Values	1589

Column	timestamp
Data Type	int64
Non-Null Count	3683
Null %	0.00%
Unique Values	3411
Min	1137179352.00
Max	1537098603.00
Mean	1320031966.82
Std Dev	172102450.44

Data Quality Metrics

movies

Metric	Value
Completeness	100.00%
Validity	100.00%
Quality Score	100.00%
Total Cells	29226
Missing Cells	0

ratings

Metric	Value
Completeness	100.00%
Validity	100.00%
Quality Score	100.00%
Total Cells	403344
Missing Cells	0

links

Metric	Value
Completeness	99.97%
Validity	100.00%
Quality Score	99.98%
Total Cells	29226
Missing Cells	8

tags

Metric	Value
Completeness	100.00%
Validity	100.00%
Quality Score	100.00%
Total Cells	14732
Missing Cells	0

Issues & Recommendations

Dataset	Issue Type	Severity	Message
movies	Data Type Mismatch	MEDIUM	Type mismatches in: ['title', 'genres']
links	Missing Values	MEDIUM	Total missing values: 8
links	Data Type Mismatch	MEDIUM	Type mismatches in: ['tmdbld']
tags	Data Type Mismatch	MEDIUM	Type mismatches in: ['tag']